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Report
GIN456: Human-Robot Interaction

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Human–Robot Interaction Project Proposal

Course: GIN456 – Human-Robot Interaction
Institution: Holy Spirit University of Kaslik (USEK)
Semester: Spring 2025-2026

1. Project Description

This project consists of deploying the Pepper humanoid robot in a private formal event where it acts as a hostess and a waitress. The robot will manage guest reception, verification, guidance, and ordering.

Deployment Context (Where)

The robot will be deployed at the entrance and inside the event hall of a private dinner or professional reception. It will interact with guests upon arrival, accompany them to their assigned tables, and take orders.

Target Audience (Who)

The intended users are invited guests attending the event. These include professionals, academics, and formal event participants who expect a smooth and organized experience.

Objectives (Why)

The main objectives of this project are:
- To design a socially appropriate and professional robot behavior.
- To implement multi-modal interaction (speech, gesture, and tablet interface).
- To manage turn-taking and structured dialogue.

System Overview

The system operates in six main stages:

1. Greet and welcome guests.
2. Enter code and show to table.
3. Show Menu, Take Order, and Billing.
4. Invite Guest to Join the Dance Floor
5. Closing and farewell.
6. Manage guest database and face recognition (which will be presented apart).

2. Motivation Behind Selecting the Topic

In large private events, managing guest reception and organization can be challenging and labor-intensive. Human staff must verify invitations, guide guests, and take initial orders, which may cause delays and inefficiencies.
Introducing Pepper as an intelligent social robot can:
- Improve organizational efficiency.
- Reduce waiting time at reception.
- Provide an innovative and engaging guest experience.
- Demonstrate the potential of robots in hospitality and service industries.

This project also allows us to explore key Human-Robot Interaction concepts such as social presence, proxemics, politeness strategies, trust, and multimodal communication.

3. Activities and Tools Needed

Hardware:
- Pepper Robot
- Event space for testing navigation and interaction

Software and Development Tools:
- NAOqi Framework
- QiChat for dialogue management
- Python for guest list verification and order processing
- Pepper Tablet Interface for menu interaction
-Web tools(HTML, CSS, JavaScript)
-Capcut for videos
-Chatgpt Pro for picture generating

Robot Activities:
- Greeting guests with appropriate gestures and tone
- Face/name recognition(done apart) and confirmation dialogue

- Guest list lookup and table assignment
- Guiding guests using the map of the event
- Displaying digital menu on tablet
- Order confirmation and structured interaction closure
-Present bill

4. Role and Work Division

- Dialogue Design & Interaction Flow: Designing structured conversations and confirmation strategies.
- Programming & Integration: Implementing QiChat and Python scripts.
- Tablet Interface & Menu Design: Developing and integrating tablet-based ordering system.
- Testing & Evaluation: Conducting user testing and analyzing user experience feedback.

5. Expected Outcomes

By the end of the project, Pepper will function as an autonomous host and service assistant in a formal private event. The project will demonstrate effective multimodal interaction, structured dialogue design, error handling strategies, and evaluation of user perception in a professional social robotics context.

Use Case 1: Greet and Welcome Guest

Pepper Robot Hostess & Waiter System

UC-1 | Greet and Welcome Guest	
Created By	Charbel Ghaya
Creation Date	April 2026
Actors	Primary Actor: Guest / Invited Person Secondary Actor: Pepper Robot (Hostess system)
Trigger	The Guest arrives at the venue entrance and Pepper detects their presence.
Description (Objectives / Goals)	The Guest wants to be welcomed by Pepper and provide a name so the interaction can continue in a personalized way.
Preconditions	PRECOND-1. Pepper is powered on and the hostess behavior is running. PRECOND-2. The greetings2 dialogue topic is loaded. PRECOND-3. The welcome image and name input page are available in the project files.
Postconditions	POSTCOND-1. The Guest has been welcomed. POSTCOND-2. The Guest name is captured either from the configured names concept or from the tablet event. POSTCOND-3. Pepper can continue to the next interaction stage.

Action Sequence — Main Success Scenario

1. The project starts and Pepper displays pics/welcometo.png.
2. Pepper says: Hello, welcome to the event.
3. Pepper explains that it will guide the user through the check-in process.
4. Pepper asks the Guest for their name.
5. The Guest says a name recognized by the dialogue or enters a name on the tablet.
6. Pepper stores the name in the $name variable.
7. Pepper welcomes the Guest by name.
8. Use case ends.

Extensions — Alternative Scenarios & Exceptions

5a. Name is entered on the tablet:
5a1. nameInput.js validates that the name field is not empty.
5a2. The tablet raises the UserNameEntered event.
5a3. Pepper receives the inputName event and returns to step 6.

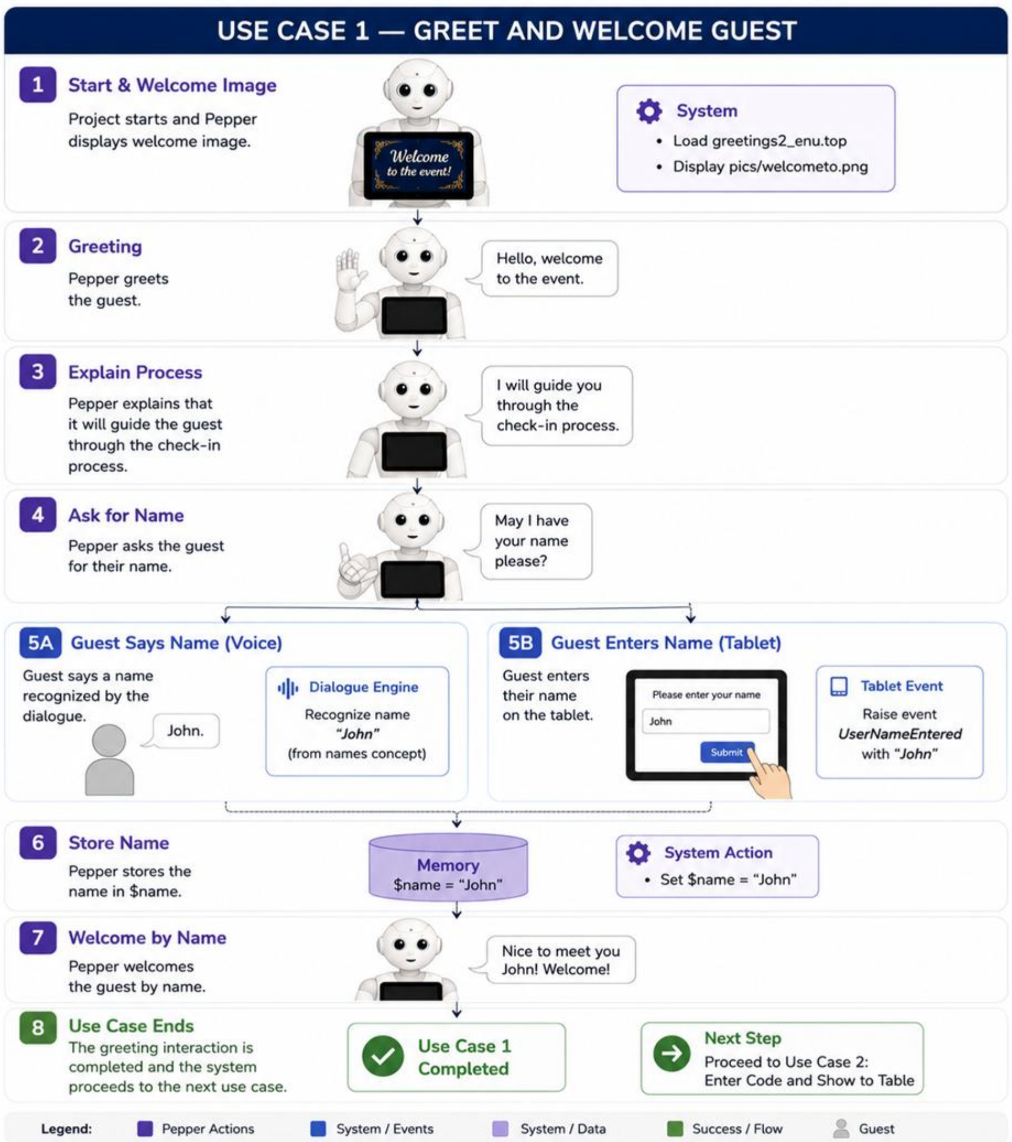
5b. Name field is empty:
5b1. The tablet displays the message Please enter your name.
5b2. The Guest enters a name and returns to step 5.

Requirements	REQ-1. Pepper must load greetings2_enu.top. REQ-2. The nameInput tablet page must validate non-empty input. REQ-3. The welcome image must be available at html/pics/welcometo.png.
Business Rules	BR-1. The interaction cannot personalize the greeting until a name is received. BR-2. The name input field cannot be submitted empty.
Priority	High - This is the first interaction with the guest.
Related Use Cases	UC-02: Enter Code and Show Table UC-04: Invite Guest to Join the Dance Floor
Assumptions	ASM-1. The guest can hear Pepper or use the tablet. ASM-2. The tablet web page is available if voice/name recognition is not used.
Open Issues	OI-1. The submitted files do not include a final guest-name database beyond the configured names concept and tablet entry.

Step-by-Step Caption Reference

Scene	Illustrated Action
Step 1	Pepper starts the interaction and displays welcometo.png.
Step 2	Pepper asks the guest for a name using speech and the tablet name input page.

Step 3	The guest provides a name; Pepper welcomes the guest by name.
Extension	If the tablet name field is empty, nameInput.js displays Please enter your name.



Use Case 2: Enter Code and Show Table

Pepper Robot Hostess & Waiter System

UC-02 | Enter Code and Show Table	
Created By	Charbel Assaker and Maria Akiki
Creation Date	April 2026
Actors	Primary Actor: Guest / Invited Person Secondary Actor: Pepper Robot and tablet webview
Trigger	Pepper asks the Guest to use the tablet to enter the invitation code.
Description (Objectives / Goals)	The Guest wants to enter the invitation code and see the assigned table on the event table map.
Preconditions	PRECOND-1. The Guest has started the Pepper interaction. PRECOND-2. The pepper_gala_webview tablet page is available. PRECOND-3. The code database in pepper_gala_webview.js contains codes 0001, 0002, 0003, and 0004 mapped to tables 1, 2, 3, and 4.
Postconditions	POSTCOND-1. A valid code is accepted. POSTCOND-2. The assigned table number is shown on the tablet. POSTCOND-3. The table map is displayed/highlighted.

Action Sequence — Main Success Scenario

1. Pepper asks the Guest to enter an invitation code on the tablet.
2. The Guest taps the four-digit code.
3. pepper_gala_webview.js checks the code against the DB object.
4. The system finds a valid code.
5. The tablet displays Welcome and Table N.
6. The selected table cell is highlighted.
7. The tablet raises the invitationCode event.
8. Pepper says thank you and asks the Guest to look for the table number on the screen.
9. Pepper displays pics/tablemap.png.
10. Use case ends.

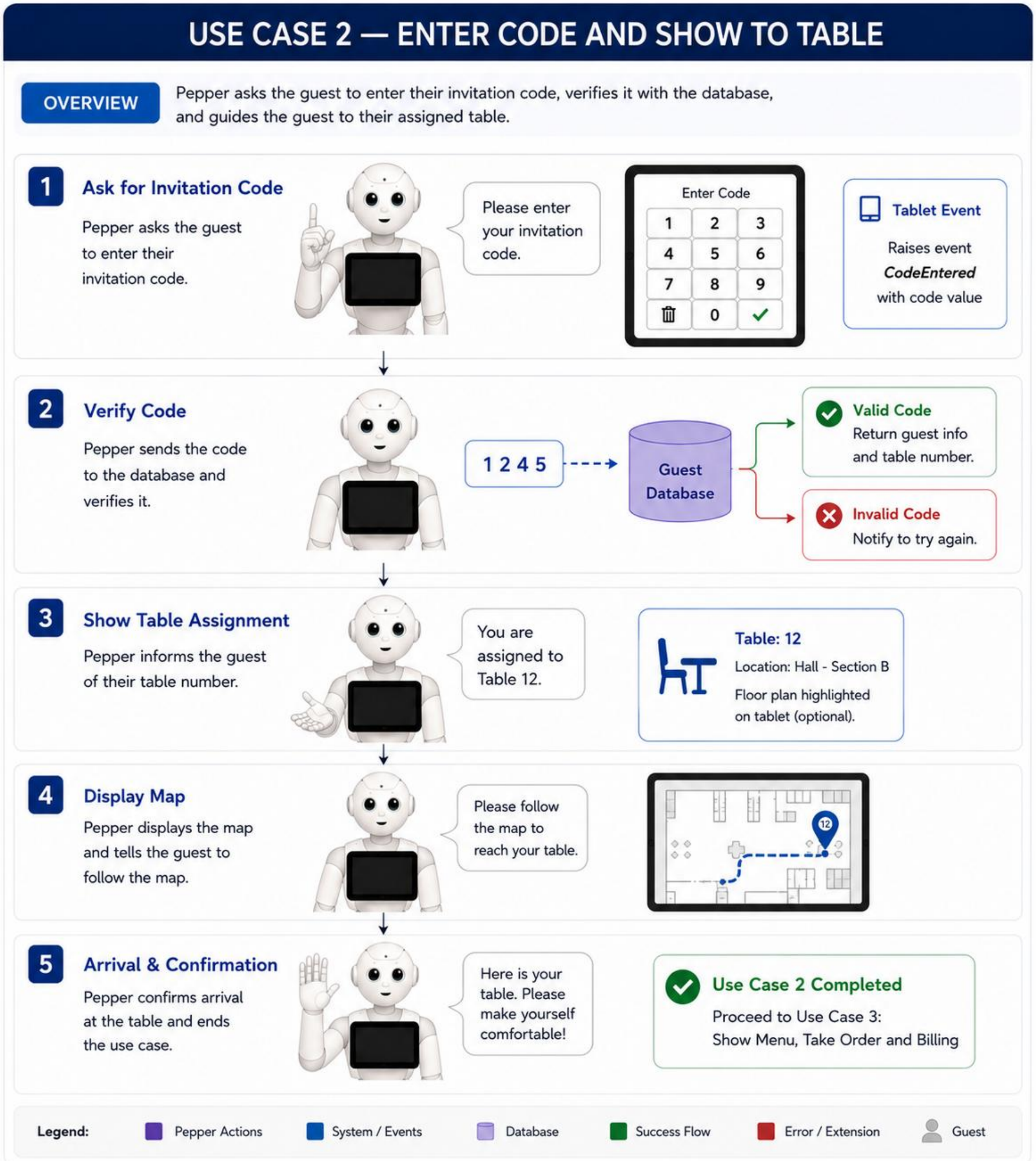
Extensions — Alternative Scenarios & Exceptions

3a. Invalid code:
3a1. The tablet displays Invalid code.
3a2. The code field is cleared.
3a3. The Guest re-enters the invitation code and returns to step 2.

Requirements	REQ-1. The tablet must support four-digit code entry. REQ-2. The project must include html/js/pepper_gala_webview.js. REQ-3. The project must include html/pics/tablemap.png.

Business Rules	BR-1. Only configured codes 0001 to 0004 are accepted in the submitted file. BR-2. Each accepted code maps to one table number
Priority	High - The table map is required before the guest proceeds to the next stage.
Related Use Cases	UC-01: Greet and Welcome Guest UC-03: Show Menu, Take Order, and Billing
Assumptions	ASM-1. The Guest has an invitation code. ASM-2. The current project supports four table numbers.

Step-by-Step Caption Reference	
Scene	**Illustrated Action**
Step 1	**Pepper asks the guest to enter the invitation code on the tablet.**
Step 2	**The guest enters a four-digit code.**
Step 3	**If valid, the tablet displays the assigned table and highlights it.**
Extension	**If invalid, the tablet alerts Invalid code and clears the code dots.**


USE CASE 2 — ENTER CODE AND SHOW TO TABLE
OVERVIEW
Pepper asks the guest to enter their invitation code, verifies it with the database, and guides the guest to their assigned table.
1
Ask for Invitation Code
Pepper asks the guest to enter their invitation code.
Please enter your invitation code.
Enter Code
1 2 3
4 5 6
7 8 9
0
Tablet Event
Raises event *CodeEntered* with code value
2
Verify Code
Pepper sends the code to the database and verifies it.
1 2 4 5
Guest Database
Valid Code
Return guest info and table number.
Invalid Code
Notify to try again.
3
Show Table Assignment
Pepper informs the guest of their table number.
You are assigned to Table 12.
Table: 12
Location: Hall - Section B
Floor plan highlighted on tablet (optional).
4
Display Map
Pepper displays the map and tells the guest to follow the map.
Please follow the map to reach your table.
12
5
Arrival & Confirmation
Pepper confirms arrival at the table and ends the use case.
Here is your table. Please make yourself comfortable!
Use Case 2 Completed
Proceed to Use Case 3: Show Menu, Take Order and Billing
Legend:
Pepper Actions
System / Events
Database
Success Flow
Error / Extension
Guest

Use Case 3: Present Menu, Take Order and Billing

Pepper Robot Hostess & Waiter System

UC-03 | Present Menu, Take Order and Billing

Created By	Charbel Assaker and Charbel Ghaya
Creation Date	April 2026
Actors	Primary Actor: Guest / Invited Person (User seated at the table) Secondary Actor: Pepper Robot (Waiter/Hostess system)
Trigger	Pepper announces that it will show the menu and activates the menu webview.
Description (Objectives / Goals)	The Guest wants to browse the menu, add items to an order, review the billing total, choose a payment method, and confirm the order.
Preconditions	PRECOND-1. The Guest has reached the menu stage of the interaction. PRECOND-2. html/pages/menu.html and html/js/menu.js are available. PRECOND-3. Menu data is loaded in menu.js.
Postconditions	POSTCOND-1. Selected items are listed in the order summary. POSTCOND-2. Subtotal, 10 percent service, and total are calculated. POSTCOND-3. A payment method is selected. POSTCOND-4. The orderConfirmed event is raised.

Action Sequence — Main Success Scenario

1. Pepper says it will show the menu.
2. The tablet displays menu categories: Appetizers, Main Dishes, and Beverages.
3. The Guest browses items.
4. The Guest taps Add To Order for one or more items.
5. menu.js stores selected items in the order array and updates quantities.
6. The Guest opens billing.
7. The tablet displays line items, subtotal, service, and total.
8. The Guest selects Cash or Card.
9. The Guest confirms billing.
10. The tablet displays the final thank-you message with total and payment method.
11. The tablet raises orderConfirmed.
12. Pepper displays pics/orderconfirm.png and says the order has been confirmed.
13. Use case ends.

Extensions — Alternative Scenarios & Exceptions

4a. Guest removes an item:
4a1. The Guest taps the minus button.
4a2. menu.js decreases quantity or removes the item.
4a3. The order totals are recalculated.

6a. Guest tries billing with empty order:
6a1. The tablet displays Please add items before going to billing.

6a2. The Guest returns to item selection.

9a. Payment method is not selected:
9a1. The tablet displays Selected: None - choose Cash or Card.
9a2. The Guest selects Cash or Card and returns to step 9.

Requirements	REQ-1. The menu page must render categories and item cards. REQ-2. The order summary must update quantities and totals. REQ-3. The billing screen must require a payment method before confirmation. REQ-4. The tablet must raise orderConfirmed when billing is confirmed.
Business Rules	BR-1. Billing cannot open if the order is empty. BR-2. Billing cannot be confirmed without Cash or Card. BR-3. Service is calculated at 10 percent in menu.js.
Priority	High - This is the main waiter function implemented in the project.
Related Use Cases	UC-02: Enter Code and Show Table UC-05: Farewell and Goodbye
Assumptions	ASM-1. The Guest can interact with the tablet. ASM-2. The menu list in menu.js is the final menu for the demo.
Open Issues	OI-1. The submitted files do not include a kitchen database or external order storage.

Step-by-Step Caption Reference

Scene	Illustrated Action
Step 1	Pepper opens the menu tablet page.
Step 2	The guest selects items from appetizers, main dishes, and beverages.
Step 3	The guest opens billing and sees subtotal, service, and total.
Step 4	The guest selects Cash or Card and confirms billing.
Step 5	Pepper confirms the order and displays orderconfirm.png.


USE CASE 3 — SHOW MENU, TAKE ORDER AND BILLING
OVERVIEW
Pepper shows the menu, takes the guest's order, confirms it, and generates the bill.
1
Display Menu
Pepper displays the menu to the guest.
MENU
Here is our menu. Please let me know your order.
Tablet Event
DisplayMenuShown event raised.
2
Take Order
Pepper records the guest's order via the tablet.
Please tell me your order.
Tablet Event
OrderEntered event raised with order details.
3
Confirm Order
Pepper shows the order summary and confirms with the guest.
You ordered:
• Item 1 x 2
• Item 2 x 1
Is this correct?
Tablet Event
OrderConfirmed event raised.
4
Generate Bill
Pepper calculates the total and generates the bill.
$ 25.00
Your total is $25.00.
System / DB Action
Calculate total and store order details and bill.
5
Display Bill
Pepper displays the bill to the guest.
TOTAL $25.00
Here is your bill. Thank you!
Tablet Event
BillShown event raised.
6
Use Case Ends
The order and billing process is completed.
Use Case 3 Completed
Next Step
Proceed to Use Case 4: Handle Special Requests
Legend:
Pepper Actions
System / Events
System / DB Actions
Success / Flow
Guest

Use Case 4: Invite Guest to Join the Dance Floor

Pepper Robot Hostess & Waiter System

UC-04 | Invite Guest to Join the Dance Floor

Created By	Maria Akiki and Arsen Eliane
Creation Date	April 2026
Actors	Primary Actor: Guest / Invited Person Secondary Actor: Pepper Robot
Trigger	Pepper reaches the dancefloor proposal in the dialogue and asks if the Guest wants to join the dance floor.
Description (Objectives / Goals)	The Guest wants to decide whether to join the dance floor. Pepper reacts to the Guest choice and either plays the dabke video or continues to farewell.
Preconditions	PRECOND-1. The interaction reaches the dancefloor proposal. PRECOND-2. The project includes html/pages/danceFloor.html. PRECOND-3. The project includes html/pics/dabke.mp4 and dabke/dabke.mp3.mpeg.
Postconditions	POSTCOND-1. If the Guest accepts, the dabke video is played. POSTCOND-2. If the Guest refuses, Pepper verbally reacts and continues. POSTCOND-3. The interaction proceeds to farewell.

Action Sequence — Main Success Scenario

1. Pepper asks: Do you want to join the dancefloor?
2. The Guest answers yes.
3. Pepper sets the video to pics/dabke.mp4.
4. Pepper starts the configured animation.
5. Pepper waits while the dance media plays.
6. Pepper proceeds to the farewell stage.
7. Use case ends.

Extensions — Alternative Scenarios & Exceptions

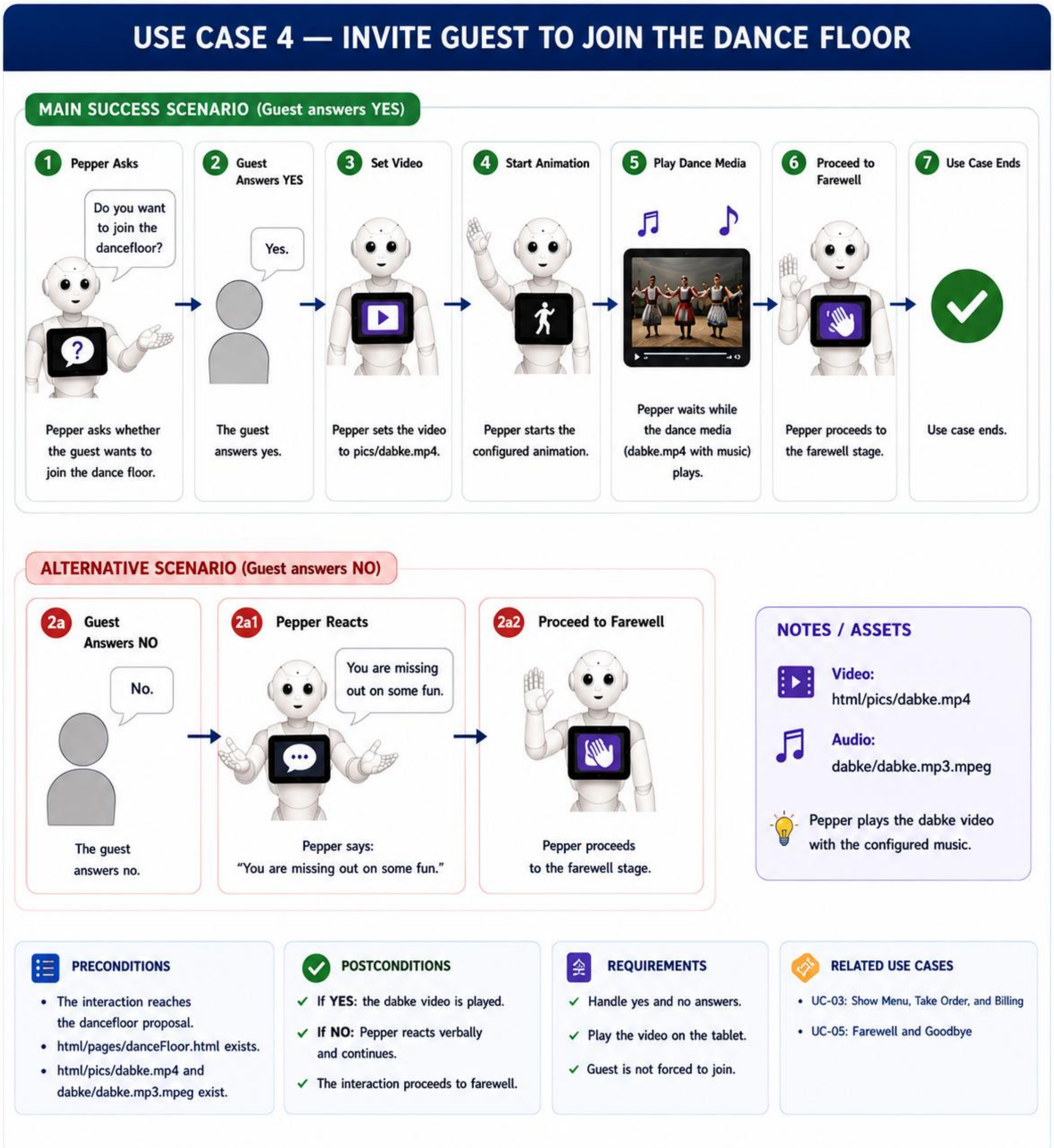
2a. Guest answers no:
2a1. Pepper says: You are missing out on some fun.
2a2. Pepper proceeds to farewell.

Requirements	REQ-1. The dialogue must handle yes and no answers. REQ-2. The dabke video asset must be available. REQ-3. Pepper must be able to play the video on the tablet.
Business Rules	BR-1. The Guest is not forced to join the dance floor. BR-2. A no answer directly continues to farewell.

Priority	Low - This is an entertainment part of the event flow.
Related Use Cases	UC-03: Show Menu, Take Order, and Billing UC-05: Farewell and Goodbye
Assumptions	ASM-1. The video file is playable on the Pepper tablet. ASM-2. The Guest can answer yes or no.
Open Issues	OI-1. The submitted files do not include a separate attendance log for dance-floor participation.

Step-by-Step Caption Reference

Scene	Illustrated Action
Step 1	**Pepper asks whether the guest wants to join the dance floor.**
Step 2	**If the guest answers yes, Pepper plays dabke.mp4.**
Step 3	**After the dance media, Pepper continues to farewell.**
Extension	**If the guest answers no, Pepper says the guest is missing out on some fun and proceeds to farewell.**


USE CASE 4 — INVITE GUEST TO JOIN THE DANCE FLOOR
MAIN SUCCESS SCENARIO (Guest answers YES)
1 Pepper Asks
Do you want to join the dancefloor?
Pepper asks whether the guest wants to join the dance floor.
2 Guest Answers YES
Yes.
The guest answers yes.
3 Set Video
Pepper sets the video to pics/dabke.mp4.
4 Start Animation
Pepper starts the configured animation.
5 Play Dance Media
Pepper waits while the dance media (dabke.mp4 with music) plays.
6 Proceed to Farewell
Pepper proceeds to the farewell stage.
7 Use Case Ends
Use case ends.
ALTERNATIVE SCENARIO (Guest answers NO)
2a Guest Answers NO
No.
The guest answers no.
2a1 Pepper Reacts
You are missing out on some fun.
Pepper says: "You are missing out on some fun."
2a2 Proceed to Farewell
Pepper proceeds to the farewell stage.
NOTES / ASSETS
Video: html/pics/dabke.mp4
Audio: dabke/dabke.mp3.mpeg
Pepper plays the dabke video with the configured music.
PRECONDITIONS
• The interaction reaches the dancefloor proposal.
• html/pages/danceFloor.html exists.
• html/pics/dabke.mp4 and dabke/dabke.mp3.mpeg exist.
POSTCONDITIONS
✓ If YES: the dabke video is played.
✓ If NO: Pepper reacts verbally and continues.
✓ The interaction proceeds to farewell.
REQUIREMENTS
✓ Handle yes and no answers.
✓ Play the video on the tablet.
✓ Guest is not forced to join.
RELATED USE CASES
• UC-03: Show Menu, Take Order, and Billing
• UC-05: Farewell and Goodbye

Use Case 5: Closing and Farewell

Pepper Robot Hostess & Waiter System

UC-05 | Closing and Farewell	
Created By	Arsen Eliane
Creation Date	April 2026
Actors	Primary Actor: Guest / Invited Person Secondary Actor: Pepper Robot and farewell feedback page
Trigger	Pepper reaches the farewell proposal after the previous interaction stage.
Description (Objectives / Goals)	The Guest wants to end the interaction. Pepper thanks the Guest and optionally collects a star rating on the tablet.
Preconditions	PRECOND-1. The Guest has completed or skipped the prior interaction stage. PRECOND-2. html/pages/farewell.html and html/js/farewell.js are available. PRECOND-3. html/pics/bye.png is available.
Postconditions	POSTCOND-1. Pepper thanks the Guest by name. POSTCOND-2. The Guest may leave a rating. POSTCOND-3. Pepper reacts with either feedbackLow or feedbackHigh response.

Action Sequence — Main Success Scenario

1. Pepper activates the farewell page.
2. Pepper says: Thank you for joining us tonight, followed by the Guest name.
3. Pepper says it hopes the Guest enjoyed the experience.
4. Pepper offers optional tablet feedback.
5. The Guest selects a star rating.
6. The Guest submits feedback.
7. farewell.js stores the rating in anonymousReviews.
8. If the rating is above 3, the tablet raises feedbackHigh.
9. Pepper displays pics/bye.png and gives a positive goodbye response.
10. Use case ends.

Extensions — Alternative Scenarios & Exceptions

5a. No rating selected:
5a1. The tablet displays Select a rating first.
5a2. The Guest selects a rating and returns to step 6.

8a. Rating is 3 or below:
8a1. The tablet raises feedbackLow.
8a2. Pepper displays pics/bye.png and apologizes that the experience was not as expected.

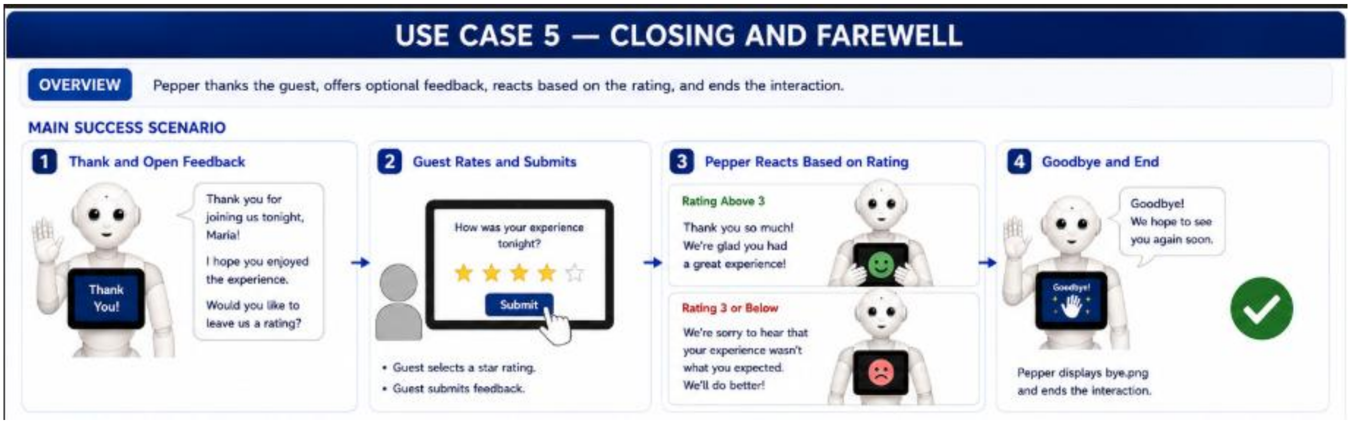
Requirements	REQ-1. The feedback page must support selecting a star rating. REQ-2. The feedback page must raise feedbackLow for ratings 3 or below and feedbackHigh

	for ratings above 3. REQ-3. The goodbye image must be available.
Business Rules	BR-1. Feedback is optional in the dialogue. BR-2. A rating cannot be submitted unless at least one star is selected.
Priority	Medium — Important for guest experience quality but not operationally critical.
Related Use Cases	UC-04: Invite Guest to Join the Dance Floor UC-01: Greet and Welcome Guest
Assumptions	ASM-1. The Guest is still using the tablet at the farewell stage. ASM-2. The rating is anonymous as implemented by the anonymousReviews array.
Open Issues	OI-1. The submitted files store reviews only in the running page memory; no permanent export file is included.

Step-by-Step Caption Reference

Scene	Illustrated Action
Step 1	Pepper thanks the guest and opens the feedback page.
Step 2	The guest chooses a star rating and submits it.
Step 3	Pepper reacts positively for high feedback or apologizes for low feedback.
Step 4	Pepper displays bye.png and ends the interaction.

Storyboard — Illustrated Scenario Walkthrough



Use Case 6: Manage Face Recognition Module(not included in the main project though)

Pepper Robot Hostess & Waiter System

UC-06 | Manage Face Recognition Module

Created By	Georges Sarkis
Creation Date	April 2026
Actors	Primary Actor: Guest / Invited Person Secondary Actor: Pepper Robot Supporting Actor: Separate face-recognition module
Trigger	The separate face-recognition module is used outside the submitted hostess Choregraphe package.
Description (Objectives / Goals)	This use case documents the separate face-recognition work identified by the group. It is not implemented inside the submitted hostess zip, so only its role in the overall project is described.
Preconditions	PRECOND-1. A separate face-recognition module exists outside the submitted zip. PRECOND-2. Pepper or the operator can use the result of that module before or during the greeting stage.
Postconditions	POSTCOND-1. The recognized identity may support the greeting/check-in stage. POSTCOND-2. The current submitted zip remains functional without this separate module.

Action Sequence — Main Success Scenario

1. The face-recognition module is started separately.
2. The Guest is presented to the recognition module.
3. The module identifies the Guest or produces a recognition result.
4. The result can be used to support UC-01.
5. Use case ends.

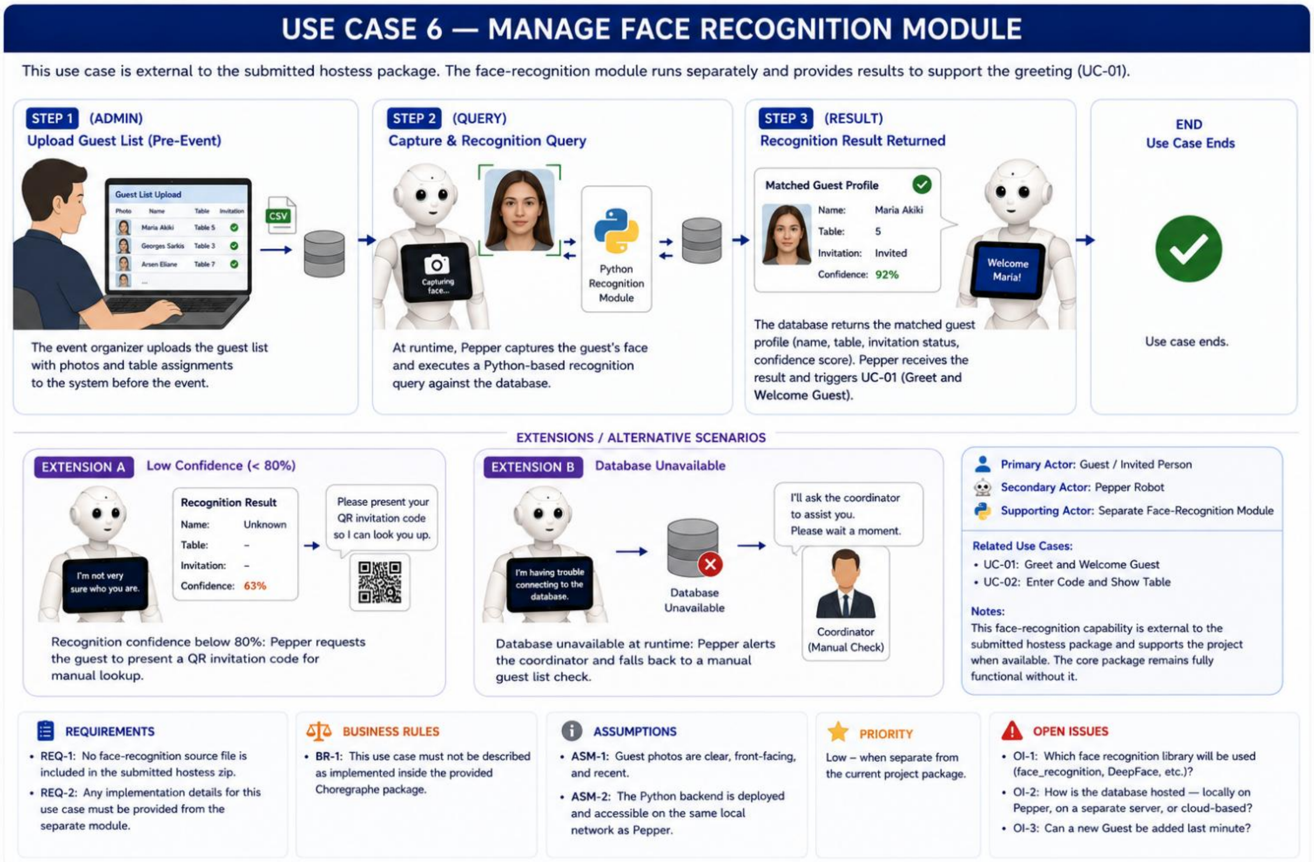
Extensions — Alternative Scenarios & Exceptions

3a. Face is not recognized:
3a1. The system cannot identify the Guest through face recognition.
3a2. The project can still use UC-01 name entry and UC-02 invitation-code entry.

Requirements	REQ-1. No face-recognition source file is included in the submitted hostess zip. REQ-2. Any implementation details for this use case must be provided from the separate module.

Business Rules	BR-1. This use case must not be described as implemented inside the provided Choregraphe package.
Priority	Low- when separate from the current project package.
Related Use Cases	UC-01: Greet and Welcome Guest UC-02: Enter Code and Show Table
Assumptions	ASM-1. Guest photos provided by the organizer are clear, front-facing, and recent. ASM-2. The Python backend is deployed and accessible on the same local network as Pepper.
Open Issues	OI-1. Which face recognition library will be used (face_recognition, DeepFace, etc.)? OI-2. How is the database hosted — locally on Pepper, on a separate server, or cloud-based? OI-3. Can a new Guest be added last minute?

Storyboard — Illustrated Scenario Walkthrough



Step-by-Step Caption Reference

Scene	Illustrated Action
Step 1 (Admin)	The event organizer uploads the guest list with photos and table assignments to the system before the event.
Step 2 (Query)	At runtime, Pepper captures the guest's face and executes a Python-based recognition query against the database.
Step 3 (Result)	The database returns the matched guest profile (name, table, invitation status, confidence score); Pepper receives the result and triggers UC-01.
Extension A	Recognition confidence below 80%: Pepper requests the guest to present a QR invitation code for manual lookup.
Extension B	Database unavailable at runtime: Pepper alerts the coordinator and falls back to a manual guest list check.

Pepper Robot Personas

Based on UC-01 to UC-06

	"Welcome me by name"	
Demographics	**Name**	Charbel Tawk
	Age	24
	Role	Guest using the greeting/name stage
	Use case focus	UC-01 Greet and Welcome Guest
	Tech comfort	High
Motivation	**Goals**	Be welcomed and continue the event interaction.
	Preferences	Clear speech prompt and a simple tablet name input fallback.
	Behavior	Responds to Pepper or uses the tablet name field.
	Personal challenges	Low patience.
Social Environment	**Background**	Professional footballer with multiple medals.
	Support needs	Non-empty name validation and clear welcome feedback.

	"Show me my table after I enter the code"	
Demographics	**Name**	Elio Khoury
	Age	39
	Role	Guest entering an invitation code
	Use case focus	UC-02 Enter Code and Show Table
	Tech comfort	Medium.
Motivation	**Goals**	Enter a valid code and see the assigned table.
	Preferences	Four-digit code entry and visible table highlight.
	Behavior	Uses the tablet keypad.
	Personal challenges	May enter an invalid code and need retry feedback since he left his glasses at home.
Social Environment	**Background**	Math instructor at SABIS.
	Support needs	Clear invalid-code message and table-map display.

	"Let me order and confirm the bill"	
Demographics	**Name**	Maya Hayeck
	Age	18
	Role	Guest ordering from the tablet menu
	Use case focus	UC-03 Show Menu, Take Order, and Billing
	Tech comfort	High, computer engineering student at MIT.
Motivation	**Goals**	Choose food/drinks and confirm the final bill.
	Preferences	Category tabs, Add To Order buttons, quantities, totals, and Cash/Card choice.
	Behavior	Adds/removes items before billing.
	Personal challenges	May judge the webview since she is a computer engineering student.
Social Environment	**Background**	Event guest using Pepper as a waiter interface.
	Support needs	

	“Invite me to the dance floor"	
Demographics	**Name**	Elissa Abboud
	Age	Not specified in project files
	Role	72
	Use case focus	UC-04 Invite Guest to Join the Dance Floor
	Tech comfort	Low
Motivation	**Goals**	Not join the dance floor due to her back pain.
	Preferences	Short yes/no prompt and entertaining media if accepted.
	Behavior	Answers yes or no.
	Personal challenges	Back pain so no dancing
Social Environment	**Background**	Retired history teacher.
	Support needs	Opts to skip and continue to farewell.

	"Let me say goodbye and leave feedback"	
Demographics	**Name**	Rita Akiki
	Age	51
	Role	Guest at the farewell stage
	Use case focus	UC-05 Farewell and Goodbye
	Tech comfort	Not specified in project files
Motivation	**Goals**	End the interaction and optionally rate the experience.
	Preferences	Fast star rating and clear response from Pepper.
	Behavior	Selects a rating or skips the feedback prompt.
	Personal challenges	Cannot submit feedback without selecting a rating.
Social Environment	**Background**	Floor manager at the supermarket.
	Support needs	FeedbackHigh/feedbackLow response and goodbye screen.

	"Recognise me"	
Demographics	**Name**	Ali Abbas
	Age	33
	Role	Getting recognized by the robot
	Use case focus	UC-06 Manage Face Recognition Module
	Tech comfort	High.
Motivation	**Goals**	Provide face-recognition support outside the submitted hostess zip.
	Preferences	Implementation details should be documented from the separate module when provided.
	Behavior	Not represented by code in the submitted zip.
	Personal challenges	The submitted zip does not include the face-recognition source files.
Social Environment	**Background**	Chemist.
	Support needs	Separate files are required to verify implementation details.

Implementation

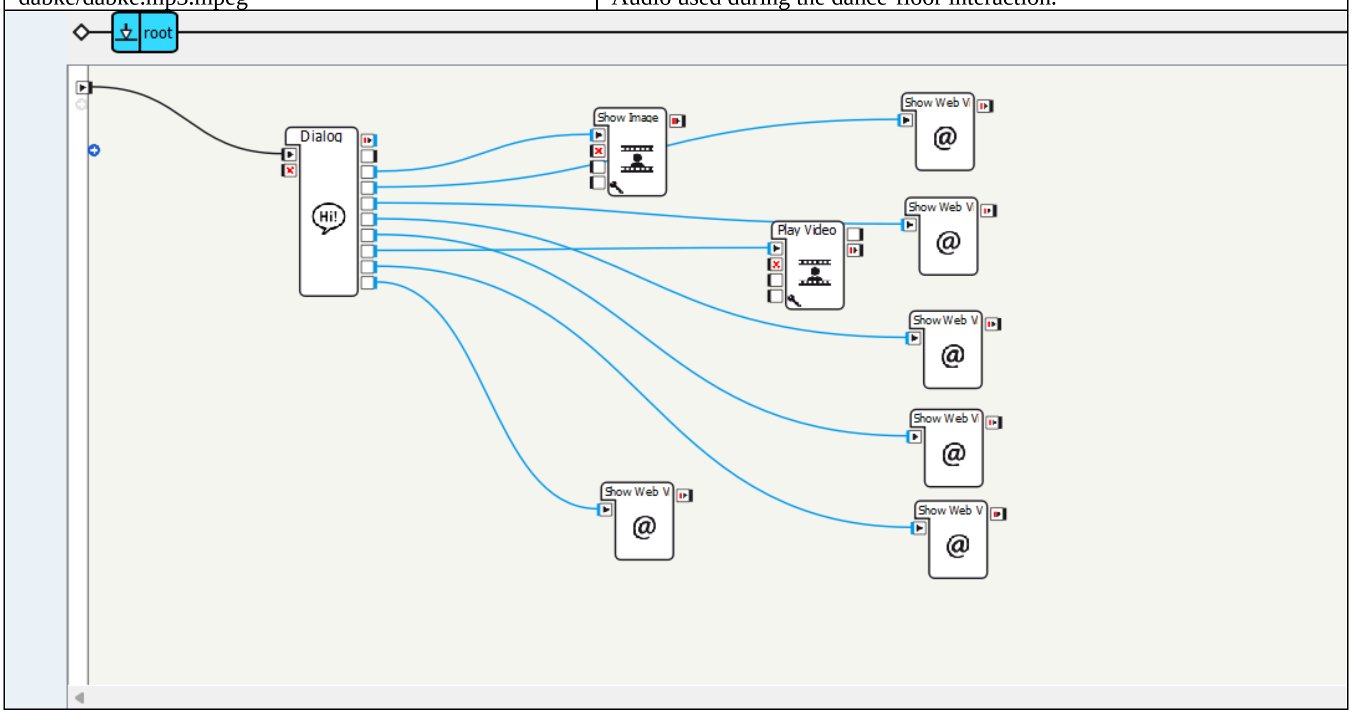
Pepper Robot Hostess & Waiter System

Overview. The system was implemented as a Choregraphe project for the Pepper robot. It combines Pepper speech, dialogue management, tablet web pages, images, video media, JavaScript events, and CSS-styled interfaces. The complete interaction guides a guest from greeting and check-in, to table display, menu ordering, dance-floor invitation, feedback, and farewell.

1. Choregraphe Project Structure

The submitted project folder is named **hostess**. It contains the Choregraphe behavior, dialogue topics, tablet pages, JavaScript logic, CSS styling, and media assets.

Folder / File	Purpose
behavior_1/behavior.xar	Main Choregraphe behavior flow.
greetings2/greetings2_enu.top	Main QiChat dialogue controlling the interaction.
html/pages/	Tablet interface pages used during the interaction.
html/js/	JavaScript files for events, menu, code entry, name input, and feedback.
html/css/	CSS files used to style tablet pages.
html/pics/	Images and videos displayed on Pepper tablet.
dabke/dabke.mp3.mpeg	Audio used during the dance-floor interaction.



2. General Interaction Flow

The interaction starts when Pepper launches the hostess behavior. Pepper welcomes the guest, asks for the guest name, asks for the invitation code, shows the table map, presents the menu, confirms the order, invites the guest to the dance floor, and finally collects feedback before saying goodbye.

```
Main dialogue file: greetings2/greetings2_enu.top

concept:(names) [elio charbel arsen georges elie maria rita] #some names may be added later on
concept:(yesAnswer) [yes Yes yeah Yeah yep Yep]
concept:(noAnswer) [no No nope Nope]
```

3. UC-01 Implementation: Greet and Welcome Guest

The greeting begins with the onStart event. Pepper displays the welcome image, starts a greeting animation, greets the guest, explains the check-in process, and asks for the guest name.

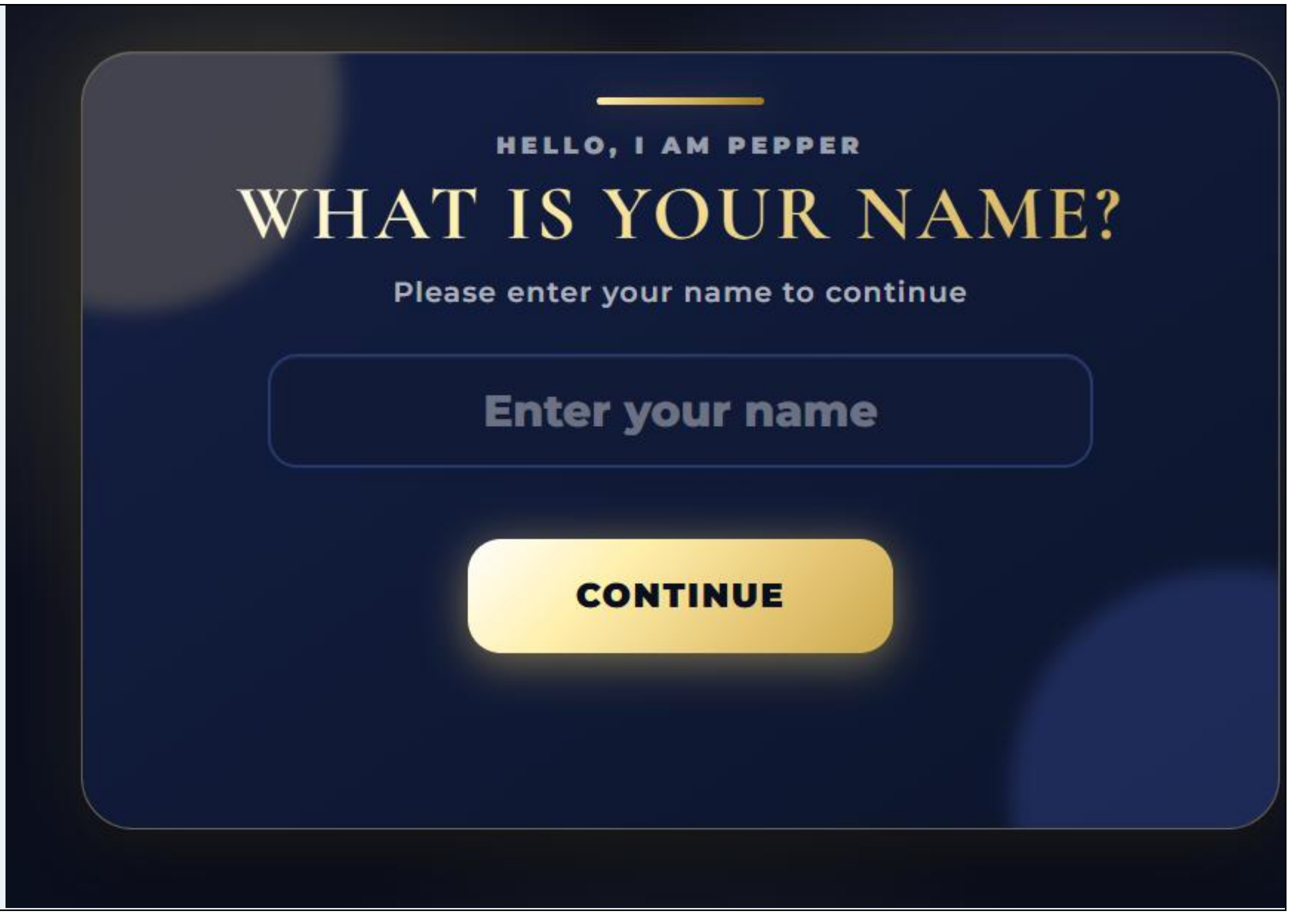
```
u:(e:onStart)
$img="pics/welcometo.png"
^start(animations/Stand/Gestures/Hey_3)
Hello, welcome to the event
```

The guest name can be received either by voice input or by tablet input. Voice input uses the names concept in QiChat. Tablet input uses:

```
html/pages/nameInput.html
html/js/nameInput.js
html/css/nameInput.css
```

Function / Event	Description
connectToPepper()	Opens a QiSession connection with Pepper.
submitName()	Reads the name typed by the guest.
showError(message)	Displays an error if the name field is empty.
clearError()	Clears the error message.
inputName	ALMemory event raised by the tablet when a valid name is submitted.

```
memory.raiseEvent("inputName", name);
u1:(e:inputName) $name=$inputName welcome $name
```



4. UC-02 Implementation: Enter Code and Show Table

The invitation-code stage uses a tablet page where the guest enters a four-digit code. The JavaScript file checks the code against a small database and displays the assigned table.

```
html/pages/pepper_gala_webview.html
html/js/pepper_gala_webview.js
html/css/pepper_gala_webview.css
```

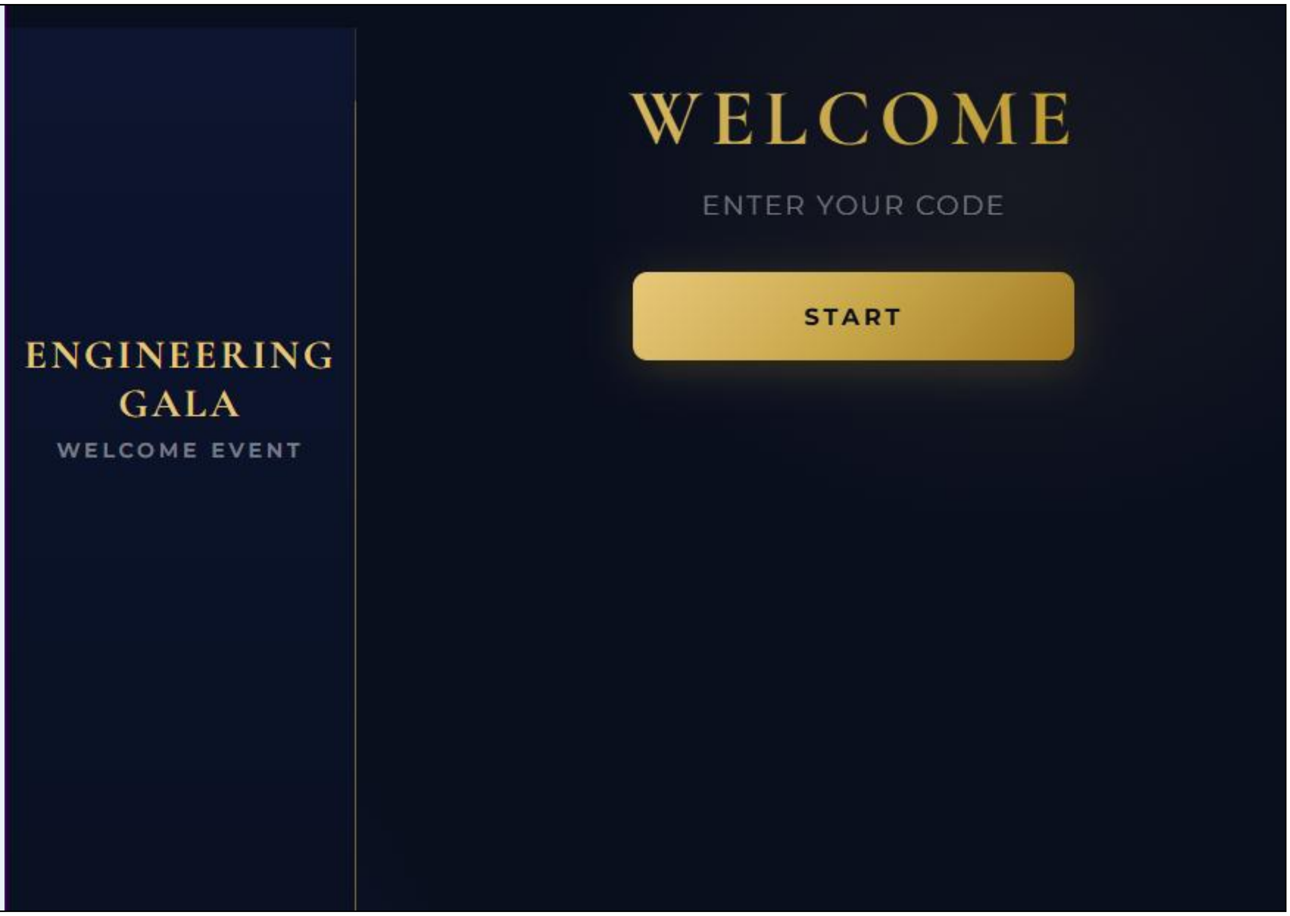
```
var DB = {
  "0001": { name: "to the event", table: 1 },
  "0002": { name: "to the event", table: 2 },
  "0003": { name: "to the event", table: 3 },
  "0004": { name: "to the event", table: 4 }
};
```

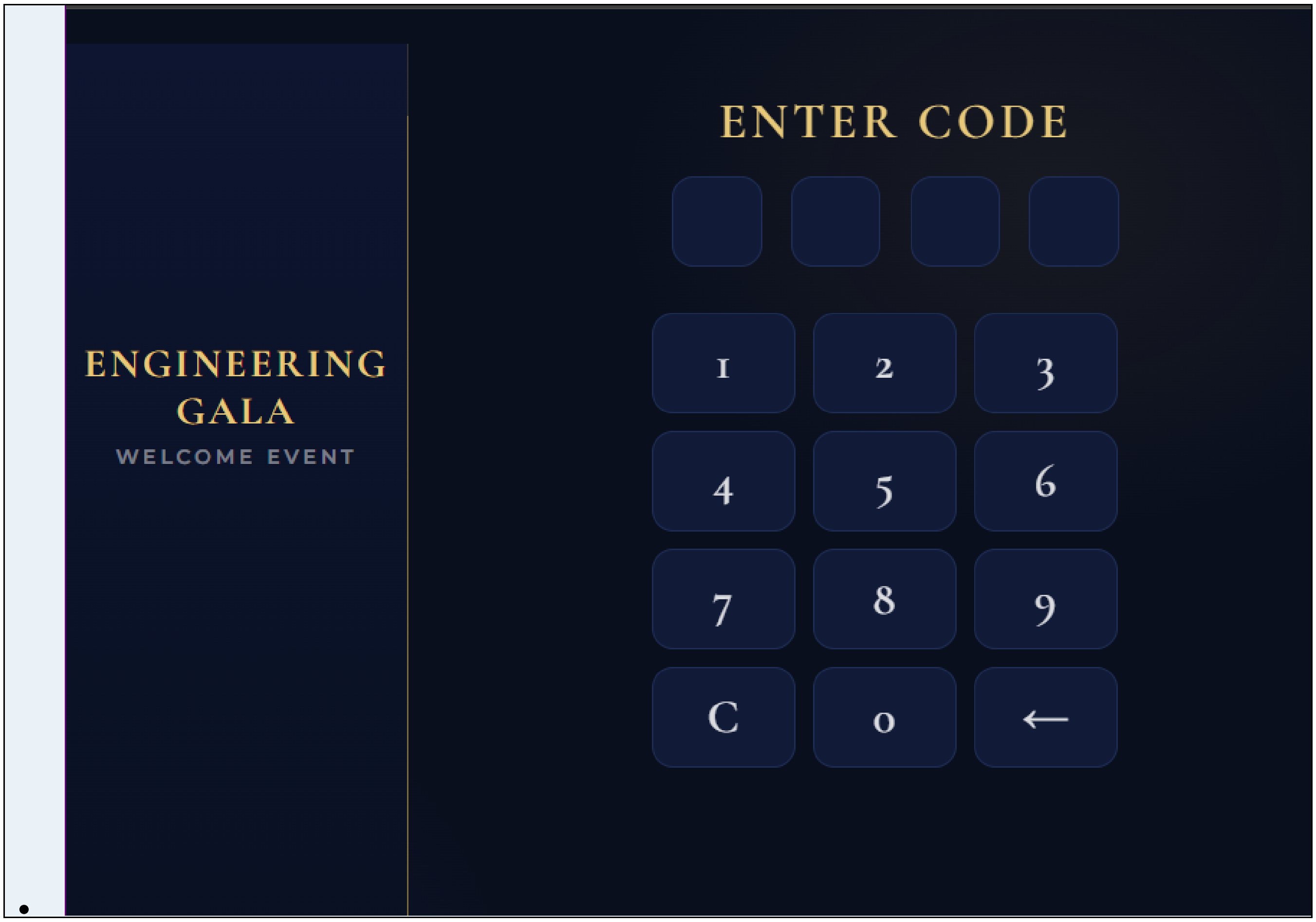
Function / Event	Description
invitationCode(n)	Adds each digit entered by the guest.
check()	Verifies the four-digit code against the DB object.

highlight(g.table)	Highlights the selected table on the tablet map.
invitationCode	Event raised after a valid code is entered.

After receiving the invitationCode event, Pepper thanks the guest and displays the table map image. Pepper does not physically guide or navigate; the guest follows the map shown on the screen.

```
$img="pics/tablemap.png"
Here is the map of the tables. The entrance is shown at the bottom.
```




ENTER CODE
1
2
3
4
5
6
7
8
9
C
0
←
ENGINEERING GALA
WELCOME EVENT

5. UC-03 Implementation: Show Menu, Take Order, and Billing

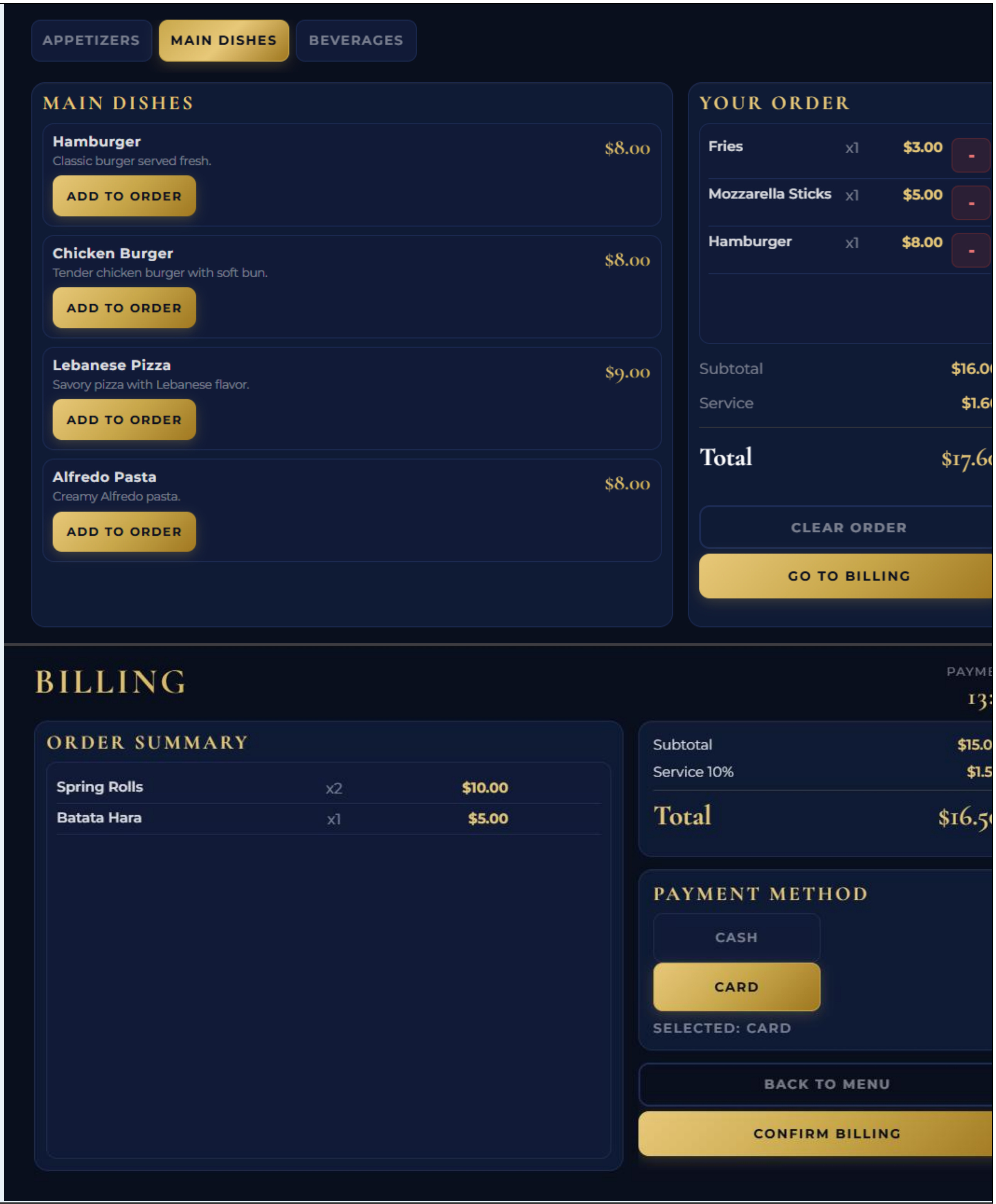
The menu and billing system is implemented as a tablet web interface. The guest browses menu categories, adds items to the order, reviews the bill, selects a payment method, and confirms the order.

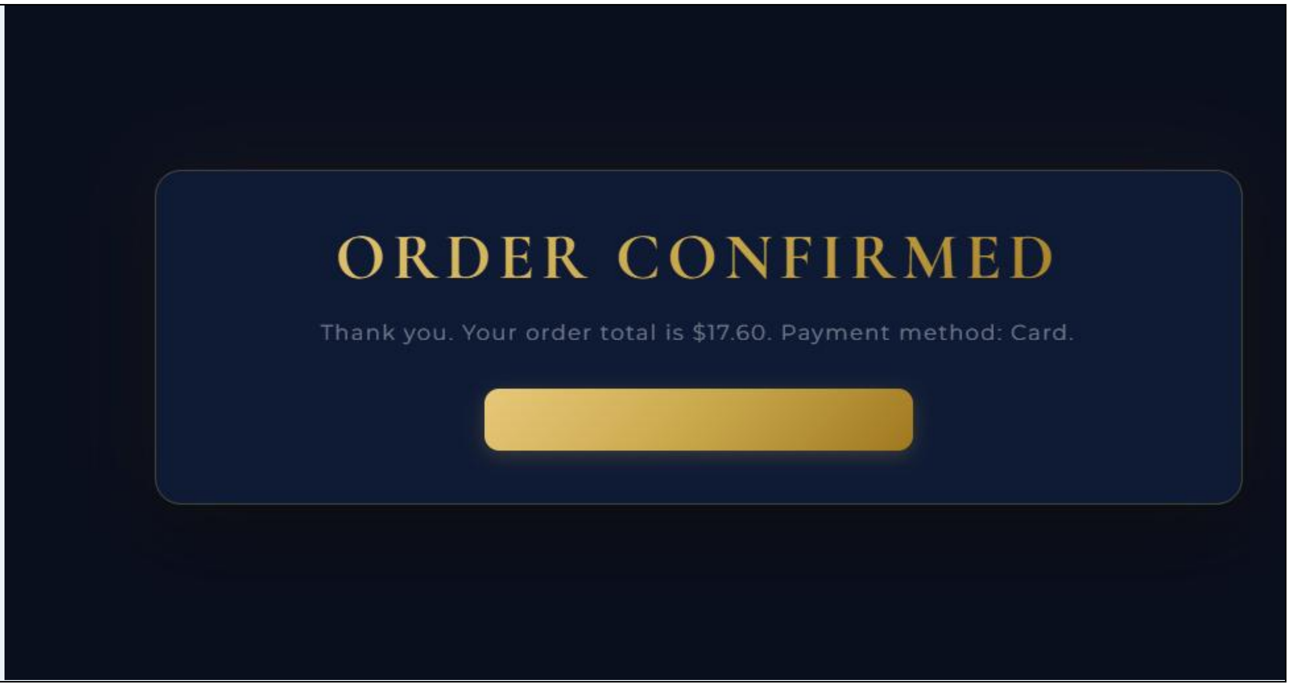
```
html/pages/menu.html
html/js/menu.js
html/css/menu.css
```

Function	Description
renderTabs()	Displays category tabs.
renderMenu()	Displays items based on the selected category.
addItem(item)	Adds an item to the current order.
removeItem(index)	Removes or decreases item quantity.
renderOrder()	Updates the order summary.
calculateTotals()	Calculates subtotal, service fee, and total.
openBilling()	Opens the billing screen.
selectPayment(method)	Saves the selected payment method: Cash or Card.
confirmBilling()	Confirms the order and raises orderConfirmed.

```
var serviceRate = 0.10;
raiseEvent("orderConfirmed", 1);
$img="pics/orderconfirm.png"
```

The system includes a 10 percent service fee. After the order is confirmed, Pepper displays the order confirmation image and continues the interaction.





6. Cooking Media Transition

After confirming the order, Pepper can display a cooking video to create a short entertainment transition before the dance-floor invitation.

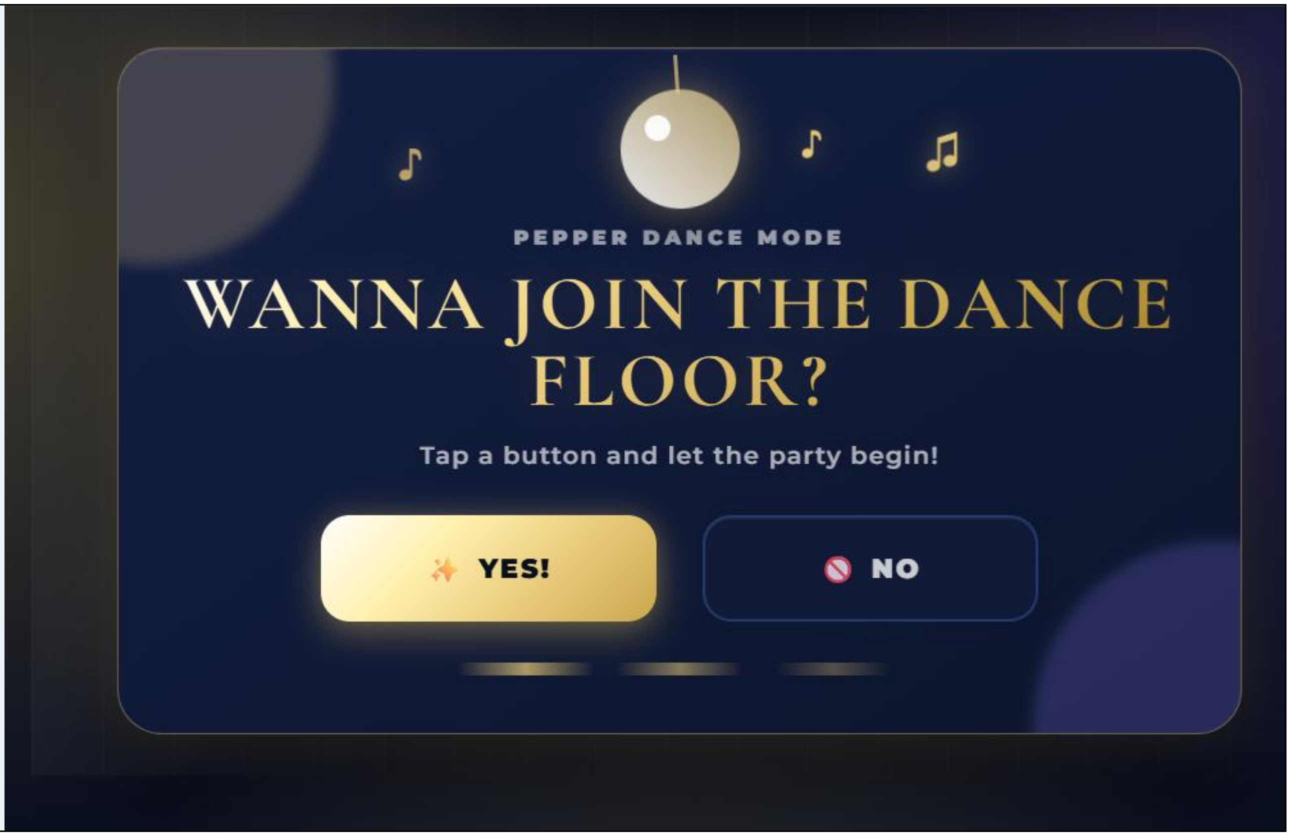
```
$video="pics/cooking.mp4"
proposal: %cook
Let's cook for you now
```

7. UC-04 Implementation: Invite Guest to Join the Dance Floor

The dance-floor feature asks the guest whether they want to join the dance floor. If the guest answers yes, Pepper plays the Dabke video and starts the configured animation. If the guest answers no, Pepper verbally reacts and proceeds to farewell.

```
html/pages/danceFloor.html
html/css/danceFloor.css
html/pics/dabke.mp4
dabke/dabke.mp3.mpeg
```

```
Do you want to join the dancefloor?
raiseConfirmationEvent("yesAnswer")
raiseConfirmationEvent("noAnswer")
$video="pics/dabke.mp4"
^start(animations/Stand/Waiting/Drink_1)
```



8. UC-05 Implementation: Closing and Farewell

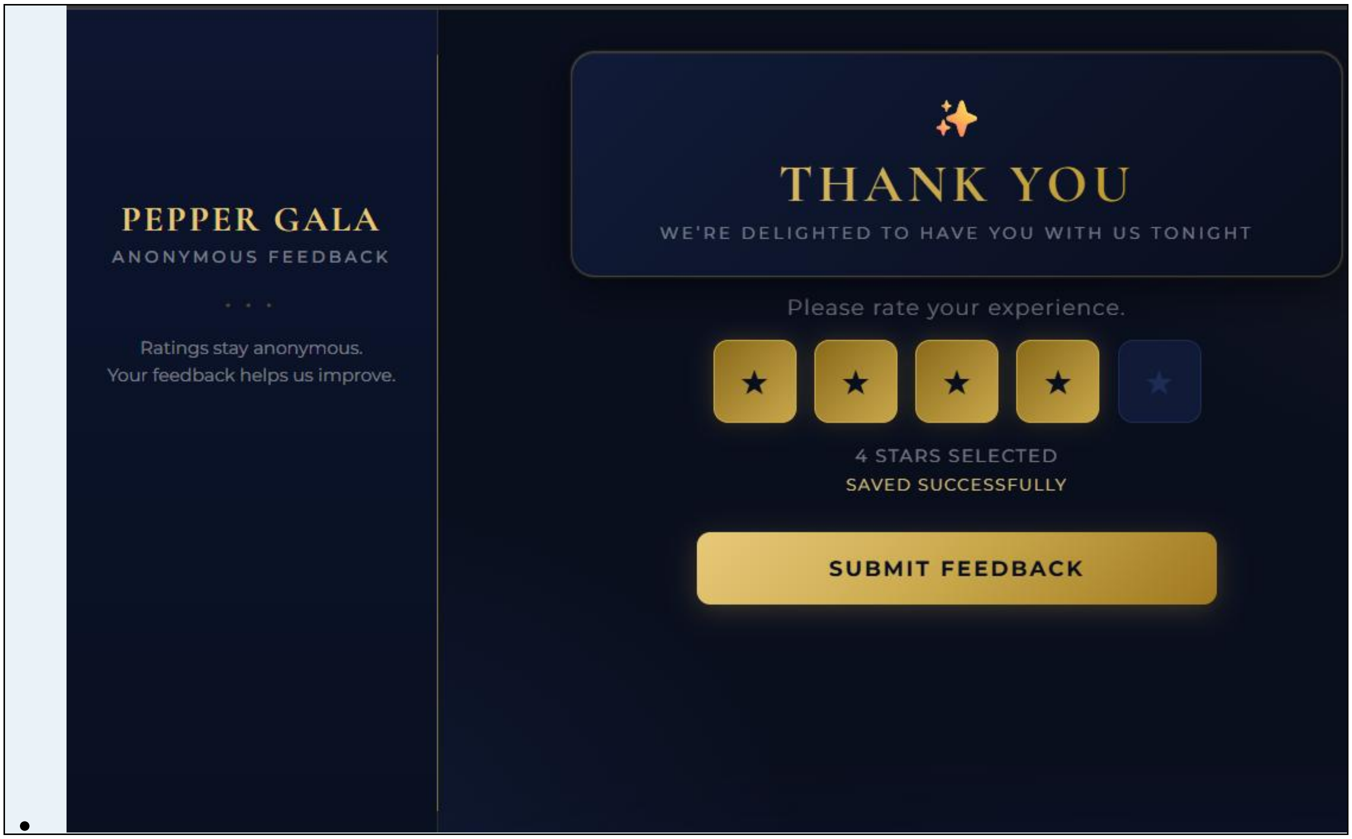
The farewell interaction thanks the guest by name, offers optional star-rating feedback on the tablet, reacts based on the rating, displays the goodbye image, and ends the interaction.

```
html/pages/farewell.html
html/js/farewell.js
html/css/farewell.css
html/pics/bye.png
```

Function / Event	Description
setRating(rating)	Stores the selected star rating.
submitGuestFeedback()	Submits the rating.
anonymousReviews	Array used to store anonymous ratings during the page session.
feedbackLow	Raised when rating is 3 or below.
feedbackHigh	Raised when rating is above 3.

```
var selectedRating = 0;
var anonymousReviews = [];
raiseMenuEvent("feedbackLow", selectedRating);
raiseMenuEvent("feedbackHigh", selectedRating);
$img="pics/bye.png"
```

If no rating is selected, the tablet displays “Select a rating first.” Ratings are anonymous and stored only in the running page memory; there is no permanent export file included.



9. UC-06 Implementation: Face Recognition Module

The face-recognition module is documented as a separate supporting module. It is not implemented inside the submitted hostess Choregraphe package. Its role is to support the greeting/check-in stage by recognizing the guest before or during UC-01.

Step	Implementation Role
1. Admin setup	The event organizer uploads the guest list with photos and table assignments before the event.
2. Recognition query	At runtime, Pepper or an external camera captures the guest face and sends it to a Python-based recognition module.
3. Result	The module returns name, table, invitation status, and confidence score.
Extension A	If confidence is below 80 percent, Pepper asks for QR invitation code/manual lookup.
Extension B	If the database is unavailable, Pepper alerts the coordinator and falls back to manual guest-list checking.

Because the source file for this module is not included in the submitted hostess zip, it should be described as external support, not as part of the current Choregraphe package.

10. Event Communication Between Tablet and Pepper

The tablet communicates with Pepper through QiSession and ALMemory events. The common helper file is:

```
html/js/qievents.js

function raiseEvent(name, value) {
  QiSession(function (session) {
    session.service("ALMemory").then(function (mem) {
      mem.raiseEvent(name, value);
    });
  });
}
```

Event	Purpose
inputName	Sends the guest name to Pepper.
invitationCode	Sends the validated invitation code.
orderConfirmed	Notifies Pepper that the order is confirmed.
yesAnswer	Guest accepts the dance-floor invitation.

noAnswer	Guest refuses the dance-floor invitation.
feedbackLow	Feedback rating is 3 or below.
feedbackHigh	Feedback rating is above 3.

11. HTML, CSS, and JavaScript Templates Used

Use Case	HTML Page	JavaScript	CSS
UC-01 Name Input	nameInput.html	nameInput.js	nameInput.css
UC-02 Invitation Code	pepper_gala_webview.html	pepper_gala_webview.js	pepper_gala_webview.css
UC-03 Menu and Billing	menu.html	menu.js	menu.css
UC-04 Dance Floor	danceFloor.html	qievents.js	danceFloor.css
UC-05 Farewell Feedback	farewell.html	farewell.js	farewell.css

12. Media Assets Used

Asset	Purpose
html/pics/welcometo.png	Welcome image during greeting.
html/pics/tablemap.png	Table map shown after invitation code.
html/pics/orderconfirm.png	Order confirmation image.
html/pics/cooking.mp4	Cooking transition video.
html/pics/dabke.mp4	Dance-floor Dabke video.
dabke/dabke.mp3.mpeg	Dabke music.
html/pics/bye.png	Goodbye image during farewell.

13. Implementation Summary

- Personalized greeting using the guest name.
- Tablet-based invitation code entry and table assignment.
- Digital menu browsing, order confirmation, billing total, and payment selection.
- Dance-floor invitation with Dabke video and yes/no response handling.
- Anonymous star-rating feedback at the end of the interaction.
- External face-recognition module documented as support for greeting/check-in.

Challenges Faced

Pepper Robot Hostess & Waiter System

Maria Akiki

The main challenge was designing the dance-floor interaction in a polite and simple way. Pepper needed to handle both yes and no answers naturally, while keeping the interaction comfortable for the guest.

Charbel Assaker

The main challenge was implementing the invitation-code and menu/billing interfaces. The system had to validate codes, show the correct table, manage selected menu items, calculate totals, and handle invalid or missing input. In addition, creating variable to take the name of the guest was challenging.

Georges Sarkis

The challenge was documenting the face-recognition module correctly because it was separate from the submitted Choregraphe package. The system also needed fallback options in case recognition failed.

Arsen Eliane

The farewell and feedback part required handling star ratings correctly. The main difficulty was making sure the guest could not submit empty feedback and that Pepper reacted differently to high and low ratings.

Charbel Ghaya

The greeting stage required both voice and tablet name input. The main challenge was making the interaction personalized while also handling cases where Pepper could not recognize the guest’s name.

Conclusion

This project showed how Pepper can be used as a hostess and waiter in a formal event. The robot welcomes guests, asks for their name, verifies their invitation code, displays the table map, shows the menu, takes orders, invites guests to the dance floor, and ends with a feedback and farewell interaction.

The project helped us apply important Human-Robot Interaction concepts such as dialogue design, tablet interaction, personalization, error handling, and user-friendly feedback. It also showed that this course was a very hands-on course based on engineering principles such as implementation, testing, debugging, integration, and improving the system after observing how it works.

Overall, the project was a valuable experience because it connected theory with practice and allowed us to build, test, and present a complete interactive robot system.